

ACCEPTED MANUSCRIPT • OPEN ACCESS

Ultrafast carrier dynamics in high-density photo-doped MoS₂ : monolayer vs multilayer

To cite this article before publication: Durga Prasad Khatua *et al* 2026 *J. Phys.: Condens. Matter* in press <https://doi.org/10.1088/1361-648X/ae82a8>

Manuscript version: Accepted Manuscript

Accepted Manuscript is “the version of the article accepted for publication including all changes made as a result of the peer review process, and which may also include the addition to the article by IOP Publishing of a header, an article ID, a cover sheet and/or an ‘Accepted Manuscript’ watermark, but excluding any other editing, typesetting or other changes made by IOP Publishing and/or its licensors”

This Accepted Manuscript is © 2026 The Author(s). Published by IOP Publishing Ltd.



As the Version of Record of this article is going to be / has been published on a gold open access basis under a CC BY 4.0 licence, this Accepted Manuscript is available for reuse under a CC BY 4.0 licence immediately.

Everyone is permitted to use all or part of the original content in this article, provided that they adhere to all the terms of the licence <https://creativecommons.org/licenses/by/4.0>

Although reasonable endeavours have been taken to obtain all necessary permissions from third parties to include their copyrighted content within this article, their full citation and copyright line may not be present in this Accepted Manuscript version. Before using any content from this article, please refer to the Version of Record on IOPscience once published for full citation and copyright details, as permissions may be required. All third party content is fully copyright protected and is not published on a gold open access basis under a CC BY licence, unless that is specifically stated in the figure caption in the Version of Record.

View the [article online](#) for updates and enhancements.

Ultrafast Carrier Dynamics in High-density Photo-doped MoS₂: Monolayer vs Multilayer

Durga Prasad Khatua^{1*}, Asha Singh², Sabina Gurung³, and J. Jayabalan^{4*}

¹Mechanical and Aerospace Engineering Department, University of California Los Angeles, Los Angeles 90095, USA.

²Nano Science Laboratory, Materials Science Section, Raja Ramanna Centre for Advanced Technology, Indore 452013, India.

³Department of Chemistry and York Biomedical Research Institute, University of York, York, YO10 5DD, UK.

⁴Faculty for physics and CENIDE, University of Duisburg-Essen, Duisburg, 47057, Germany.

*Address correspondence to: dkhatua@physics.ucla.edu, quantumbalan@gmail.com

Abstract

Monolayer and multilayer MoS₂ are extremely fascinating materials for optoelectronics, particularly in scenarios involving high excitation carrier densities. Consequently, it is essential to understand the various processes and their associated time scales in monolayer and multilayer two-dimensional materials under such high excitation densities. Here, We demonstrate that carrier dynamics in monolayer and multilayer MoS₂ exhibit distinctly different behaviors above the Mott density, with monolayers showing faster carrier relaxation and enhanced many-body interactions compared to multilayers. Despite the similarity in band structure near the *K*-point and the formation of A-excitons, pronounced differences in the transient response are observed, highlighting the influence of the overall band structure. Exciton dissociation, bandgap renormalization, and intervalley relaxation play a consequential role in dictating the ultrafast transient properties of these samples. The study in this paper provides a detailed understanding of the fundamental optoelectronic properties of the two-dimensional MoS₂ at high excitation densities, paving the way for its potential applications in various photonic and optoelectronic domains.

1 INTRODUCTION

Due to the increasing demand for scalable optoelectronic applications, the exploration of novel materials has led to the emergence of transition-metal dichalcogenides, a class of two-dimensional layered materials. Among them, molybdenum disulfide (MoS₂) is a highly studied material because of its

tunable optoelectronic properties, which can be controlled by varying the number of layers. In particular, MoS₂ shows a transition from indirect to direct band gap as it is thinned from bulk to single-layer form [1, 2]. Due to the direct band gap nature, the monolayer (MoL) MoS₂, possesses several physical properties that make it suitable for a wide range of applications, including optoelectronics, valleytronics, and quantum technologies [3–5]. On the other hand, bulk and multilayer (MuL) MoS₂ find their applications in transistors, photodetectors, solar cells, flexible electronics, and gas and biosensors, among others [6–10].

Regardless of the number of layers, MoS₂ exhibits three types of excitons in the visible region: A, B, and C [1, 2]. In addition, owing to the significant contribution of the *d*-orbital electrons from Mo-atoms, MoS₂ exhibits high spin-orbit coupling at the *K*-point of the Brillouin zone, which is strong enough to lift the degeneracy in both the valence and conduction bands [11]. In the monolayer form, quantum confinement and enhanced Coulomb interactions between charge carriers become significant, resulting in tightly bound excitons with binding energies of approximately ~ 0.5 eV [12–14]. By comparison, the exciton binding energy in bulk MoS₂ is much lower, around 45 meV [15, 16], which is still larger than that of typical bulk semiconductors such as Si, Ge, InAs, and GaAs. Moreover, monolayer MoS₂ exhibits two nonequivalent yet degenerate valleys at the *K*-points as a consequence of broken inversion symmetry. In contrast, bulk MoS₂ possesses inversion symmetry, rendering the bands near the *K* points equivalent, while the spin and optical transition properties remain similar to those of the MoL [1, 2, 15].

The application of MuL and MoL MoS₂ in optoelectronic and sensor devices necessitates a comprehensive understanding of their fundamental carrier dynamics. Investigating carrier relaxation dynamics offers valuable insight into the transient optical and electronic properties of these materials. Several research groups have studied the carrier dynamics in MoS₂ earlier to understand the carrier behavior with different excitation and probing conditions [17–27]. Until now, research has primarily focused on the carrier dynamics at low excitation densities, where perturbations in the properties due to high excitation conditions are kept to a minimum. However, several MoS₂-based applications demand handling of high excitation densities, such as optical parametric amplifier [28], modulator [29], and photodetectors [30]. Consequently, it is essential to understand the carrier dynamics at such high photo-excitation conditions.

In this work, we present the results of the carrier dynamics studied in MoL and MuL MoS₂ flakes under varying nominal photoexcited carrier densities per layer (NPL), ranging from $0.66 \times 10^{14} \text{ cm}^{-2}$ to $4.16 \times 10^{14} \text{ cm}^{-2}$ at A-exciton energy, which is above the Mott density of a MoL MoS₂ [31]. Atomic Force Microscope (AFM) measurements were performed to determine the number of layers in the selected MoS₂ flakes. The thickness of the MuL on which the carrier dynamics measurements were reported is 8.9 nm, which corresponds to about 11 layers of MoS₂. As previously reported, an increase in the number of layers leads to a gradual change in the optical property of the MoS₂ [1, 2, 32]. However, the properties remain almost unchanged if the number of layers goes beyond six [1, 2, 32]. Thus, the present sample with 11 layers can be treated as a bulk MoS₂. By investigating the transient carrier behavior in both of these samples, we present a comparison of the carrier relaxation pathways and their associated time constants. We find that the Auger and defect-assisted Auger processes play a crucial role in determining carrier relaxation in MoL, whereas an additional mech-

anism, intravalley scattering, dominates in MuL. In MoL, intervalley scattering is not prominent, as carriers are excited at the lowest transition energy, which corresponds to the A-exciton. Furthermore, the peak heights of the transient signal show a distinct saturation behavior in MuL while it remains linear in MoL MoS₂. The primary objective of the present study is to directly compare the two limiting dimensional regimes, monolayer and multilayer, in order to clearly elucidate the influence of reduced dimensionality on carrier dynamics at photoexcitation densities above the Mott density.

2 RESULTS AND DISCUSSION

The samples that are used in the ultrafast carrier dynamic measurements are procured from 2DLayers, USA, and are grown on sapphire substrate in the chemical vapor deposition (CVD) method [33]. Figure 1(a) shows the large area absorption spectrum of the flakes, which has two distinct peaks in the lower energy region corresponding to A and B-excitons [2, 18, 34]. These peaks can be well fitted with Lorentzian functions, centered at 1.83 eV and 1.98 eV with spectral widths of 0.11 eV and 0.18 eV, respectively. In addition, the absorption spectrum of the monolayer (MoL) sample used for the transient studies (Fig. S3) shows the A- and B-exciton peaks at 1.83 eV and 1.98 eV, respectively. Microscopic optical images of the sample reveal the presence of both well-separated and partially connected MoS₂ flakes. The observed transmission contrast indicates that, although the sample consists of flakes with different thicknesses, each individual flake exhibits a constant height. Subsequent atomic force microscopy (AFM) measurements confirm that most flakes are monolayers, along with a few multilayer flakes. The images of the specific MuL and MoL MoS₂ flakes used for the pump-probe measurements are shown in Fig.1(b) and Fig.1(c), respectively [35, 36]. These optical images were acquired using a home-built microscope integrated into the pump-probe setup, enabling both the identification and investigation of the selected flakes. The dotted circles in the images indicate the specific locations where transient measurements were performed in transmission mode. The thickness of the flakes was determined by performing AFM topography scans at the edges of these flakes. AFM topography scans of the MoL and MuL flakes on which the transient measurement results presented in this work are shown in Figs.2(a) and 2(c), respectively. Furthermore, the corresponding height profiles across the edges of the MoL and MuL flakes are shown in Figs.2(b) and 2(d), respectively. From the height differences at the edge of the flakes, the thicknesses of the MoL and MuL flakes are estimated to be nearly 0.8 nm and 8.9 nm, respectively. Further, the Raman spectrum (Fig. S2) confirms the single layer MoS₂. Based on the thickness of a single layer, the MuL flake is expected to be composed of 11 mono layers [1, 2, 37]. The in-plane dimensions of the MuL and MoL flakes are $\sim 35 \mu\text{m}$ and $100 \mu\text{m}$, respectively.

All transient data presented herein were acquired using a home-built, degenerate pump-probe configuration tuned to the A-exciton transition energy (Fig. S1). A more detailed description of the experimental setup is provided in the Supporting Information (SI). In this article, all the measurements are performed in the pump fluence range, 1.2 mJ cm^{-2} to 7.5 mJ cm^{-2} which corresponds to an NPL of $6.6 \times 10^{13} \text{ cm}^{-2}$ to $4.1 \times 10^{14} \text{ cm}^{-2}$ at the A-exciton wavelength [35–37]. The NPL is

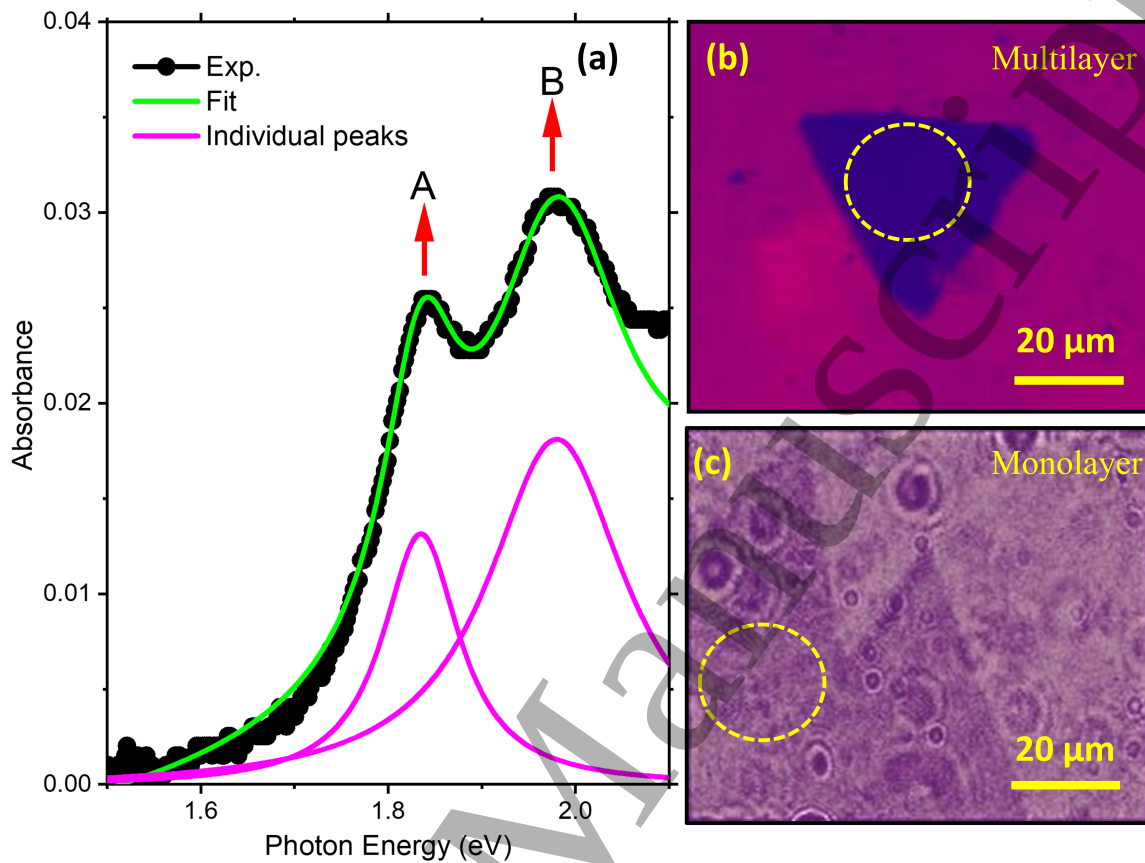


Figure 1: Optical characterization of the MuL and MoL MoS₂ flakes. (a) Large area averaged absorbance spectrum of the MoS₂ flakes. The solid green line represents the best fit to the data using two Lorentzian peaks shown as solid magenta lines. Optical image of the MuL MoS₂ flake (b), and MoL MoS₂ flake (c), acquired using a microscope. Dotted circles indicated the region where carrier dynamics measurements were performed. Optical image of the MoL MoS₂ in panel (c) is adapted from our previous work, Ref. [37]. Copyright © IOP Publishing. Adapted with permission under the IOP Authors Rights Policy.

calculated using a standard conversion relation, $N \propto \frac{E}{E_{ph}} \frac{A}{\pi r_{FWHM}^2}$ with N as number of photons, E as the laser pump pulse energy, A as absorbance, E_{ph} as energy of a single photon at the pump wavelength, and r_{FWHM} as the beam radius at full width half maximum on the sample, measured using a CCD camera. In this calculation, the absorption across the MuL is considered uniform across the film to first order, as the optical decay over such thickness is weak due to the higher optical penetration depth of ~ 81.3 nm as calculated from the absorption coefficient [38]. Since measurements were carried out at such high excitation densities, it is essential to ensure that there are no permanent modifications or damage to the flakes. Pan *et al.* studied structural modifications in MuL MoS₂ and found that the samples remained intact up to a pump fluence of 150 mJ cm^{-2} [23]. When they increased the fluence up to 400 mJ cm^{-2} , nanoridges and nanocracks were formed in the

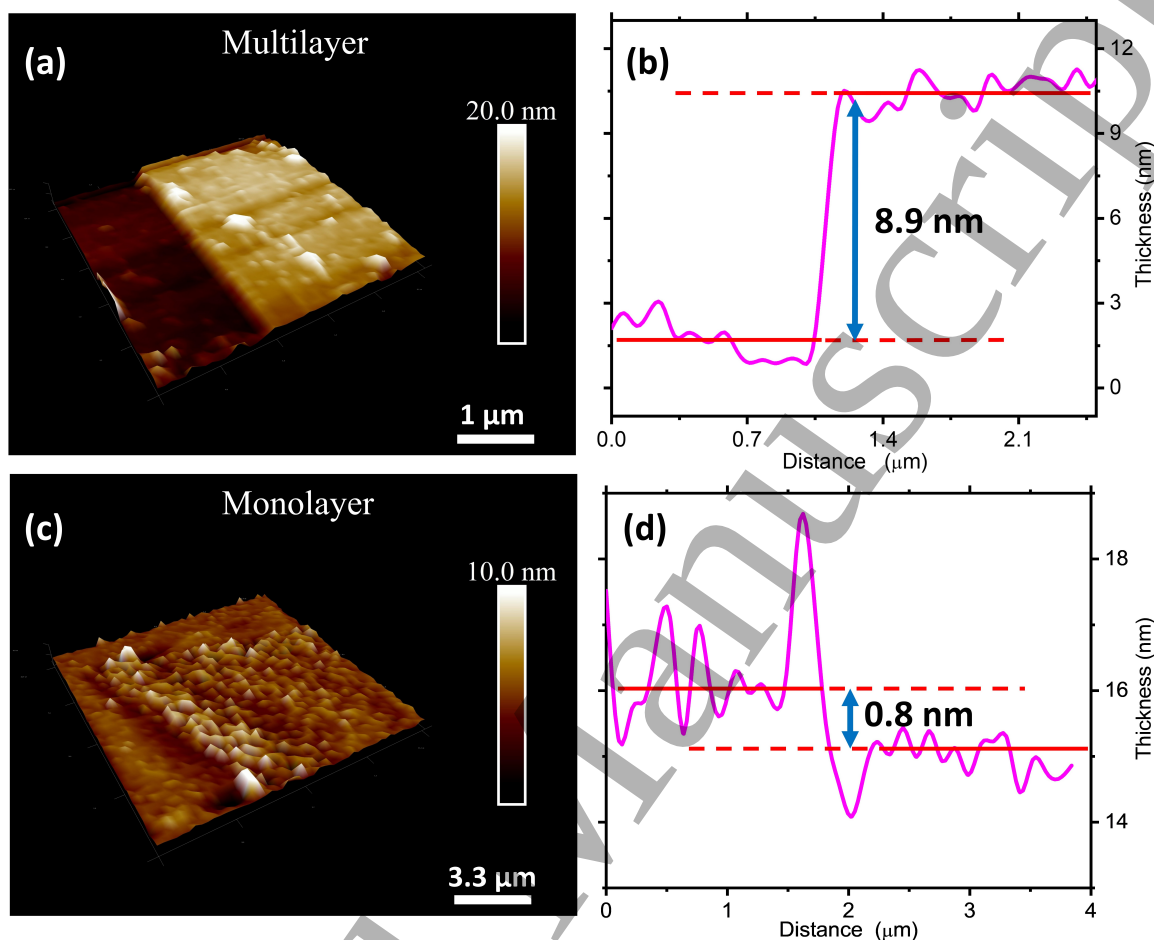


Figure 2: Panels (a) and (b): AFM topographic scan and the corresponding height profile extracted along a line perpendicular to an edge of the MuL flake, respectively. Panels (c) and (d): are the same but for the MoL flake. Panel (d) is adapted from our previous work, Ref. [37]. Copyright © IOP Publishing. Adapted with permission under the IOP Authors Rights Policy.

flakes. We have performed AFM measurements before and after our pump-probe measurements, revealing no structural changes in either MoL or MuL flakes. This indicates that the maximum fluence used in the pump-probe measurement (10 mJ cm^{-2}) is small enough to damage the flakes. Further, we have also checked that the MoL is stable even up to 50 mJ cm^{-2} [36, 37].

The fluence-dependent transient transmission signal ($\Delta T/T$) of the MuL MoS₂ flake is shown in Fig. 3(a). At low pump fluences, the transmission of the sample reduces with the arrival of the pump pulse and attains a negative peak near the end of the pump pulse. This change in the transmission then starts recovering as the time delay between the pump and probe increases and returns to its original unperturbed transmission state by about 5 ps. As the pump fluence increases, the magnitude of the peak change in the transmission also increases. At higher fluence, the recovery of transmission was found to be faster than that at lower pump fluences. Furthermore, when pump

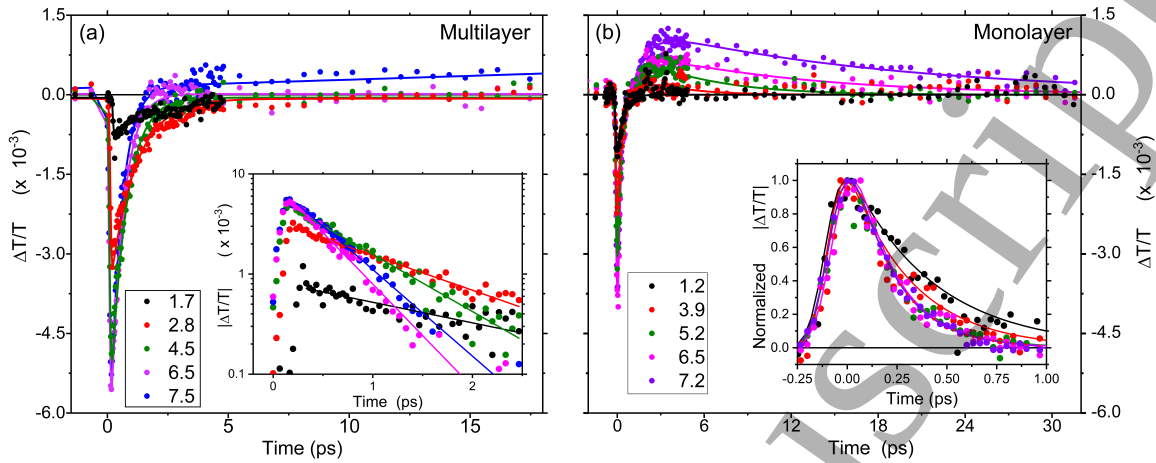


Figure 3: Fluence-dependent transient transmission signals recorded at a pump-probe wavelength of 672 nm: (a) for MuL MoS₂ flake (inset: same data on a shorter time scale is shown with a logarithmic y-axis), and (b) for MoL MoS₂ flake (inset: same data normalized and zoomed around zero delay time). The legend indicates the pump fluence in mJcm⁻².

fluence is < 6.5 mJcm⁻², the transient signal recovers completely by about 5 ps. However, as the fluence increases beyond 6.5 mJcm⁻², $\Delta T/T$ was found to change its sign from negative to positive after nearly 4 ps. The inset of Fig.3(a) shows the same transient signal of the MuL MoS₂ flake over the first few picoseconds, and the y-axis is plotted on a logarithmic scale. The solid lines in the inset plot are the linear fit to the logarithm of $|\Delta T/T|$ in 0.3 ps to 2.5 ps range. The linear relationship between the $\log(|\Delta T/T|)$ and decay time with a negative slope indicates that the $\Delta T/T$ shows a single exponential decay. From the slope of this linear fit, the decay time of $\Delta T/T$ can be obtained.

In Fig.3(b), the transient transmission signal of the MoL MoS₂ flake is shown for the same pump fluence range as that used for the MuL flake. Similar to the MuL sample, the $\Delta T/T$ signal of the MoL MoS₂ also starts decreasing with the arrival of the pump pulse reaching a negative peak. This negative change starts recovering with increasing delay time. As the pump fluence is increased, the magnitude of the maximum change in $\Delta T/T$ also increases. At lower pump fluences, the transient signal recovers completely within the first 2 ps. However, at higher pump fluences, $\Delta T/T$ changes sign from negative to positive. This positive signal builds up slowly and attains a positive maximum by about 3 ps. In the inset of Fig.3(b), the normalized $|\Delta T/T|$ of MoL MoS₂ flake is shown up to one picosecond delay between pump and probe. Clearly, the decay time of $|\Delta T/T|$ decreases with the increase in pump fluence. In addition, to assess any possible laser-induced heating effects, we measured the transient response over longer pump-probe delay times for both MoL and MuL samples, and also performed the same measurements on the sapphire substrate to evaluate its contribution to the transient signal (Fig. S4). No evidence of laser heating or substrate-induced effects was observed (see SI for a more detailed discussion).

Let us now compare the behavior of the MuL flake with that of MoL flake. Figure 4 shows the variation of the magnitude of the initial negative peak height of $\Delta T/T$ ($|\Delta T/T|_{Peak}$) with pump fluence for the MuL MoS₂ (red dots) and MoL MoS₂ (blue dots) samples. In the case of MuL MoS₂,

the $|\Delta T/T|_{Peak}$ increases with pump fluence, but at higher fluence, the peak shows signature of saturation.

At higher excitation intensities it is possible to saturate the electronic or excitonic transitions by filling a substantial fraction of the higher energy state and also by the depletion of the initial state. This leads to a reduction in the absorption under higher excitation conditions. In this regime, the intensity-dependent absorption coefficient can be expressed as,

$$\alpha = \frac{\alpha_0}{1 + I/I_S}, \quad (1)$$

where I and I_S are the intensity of excitation and saturation intensity of the material. In the experiment, both the temporal pulse width and spot size of the pump pulse are kept constant; therefore, the fluence (F) is directly proportional to the peak intensity of the pulse. The peak transient transmission is related to the change in absorption in the presence of pump by $\Delta T/T = -\Delta\alpha d$. Here, d is the thickness of the sample and $\Delta\alpha$ represents the pump-induced change in the absorption. Substituting α from Eq.1, the peak change in the transient transmission can be expressed as,

$$\left| \frac{\Delta T}{T} \right|_{Peak} = \frac{(\alpha_0 d) F}{F + F_S}, \quad (2)$$

where F denotes the fluence used in the experiment, and F_S is the saturation fluence. The above equation was used to fit the experimentally measured data for the MuL sample and is represented as a solid red curve in Fig.4. In contrast, the variation of the initial peak height $|\Delta T/T|_{Peak}$ for the MoL sample shows a linear dependence on the pump fluence (Fig.4). The solid blue line shows the linear fit to the $|\Delta T/T|_{Peak}$ data for MoL MoS₂.

Having discussed about the initial negative peak change in the $\Delta T/T$, let us now compare the associated recovery time, or equivalently the decay time of $|\Delta T/T|$, τ_s , occurring at the initial short time scale. For the MuL case, the decay time τ_s at each fluence is extracted by fitting the transient data in the range of 0.3 ps to 2.5 ps to a single exponential function. The obtained best-fit decay time for various fluences is plotted in Fig.5. The fluence dependence of τ_s is well described by the empirical relation $\tau = A/F^2 + B$, where, A and B are fitting constants. The corresponding best fit is also shown in Fig.5. The observed decrease in relation time τ_{MuL} with $1/F^2$ suggests the involvement of nonlinear relaxation processes such as Auger recombination, carrier-carrier scattering, or exciton-exciton annihilation. In this case, the presence of higher number of excited carriers enhances the relaxation dynamics via collective effect as the threshold is being crossed.

When compared to the MuL flake, the transient transmission of MoL flakes shows more complex dynamical behavior. Following the recovery of the initial negative peak, the $\Delta T/T$ becomes positive, which subsequently recovers over much longer time scales at later times (Fig.3(b)). Hence, extracting the correct recovery time in MoL is more complicated compared to MuL. A detailed discussion on the fluence dependence of the MoL flake can be found in our earlier publication [35]. However, for the sake of completeness, in short, we discuss the time dependence of the MoL sample here. To obtain the fast recovery time of the MoL case, τ_s , we fit the experimental data to an empirical function.

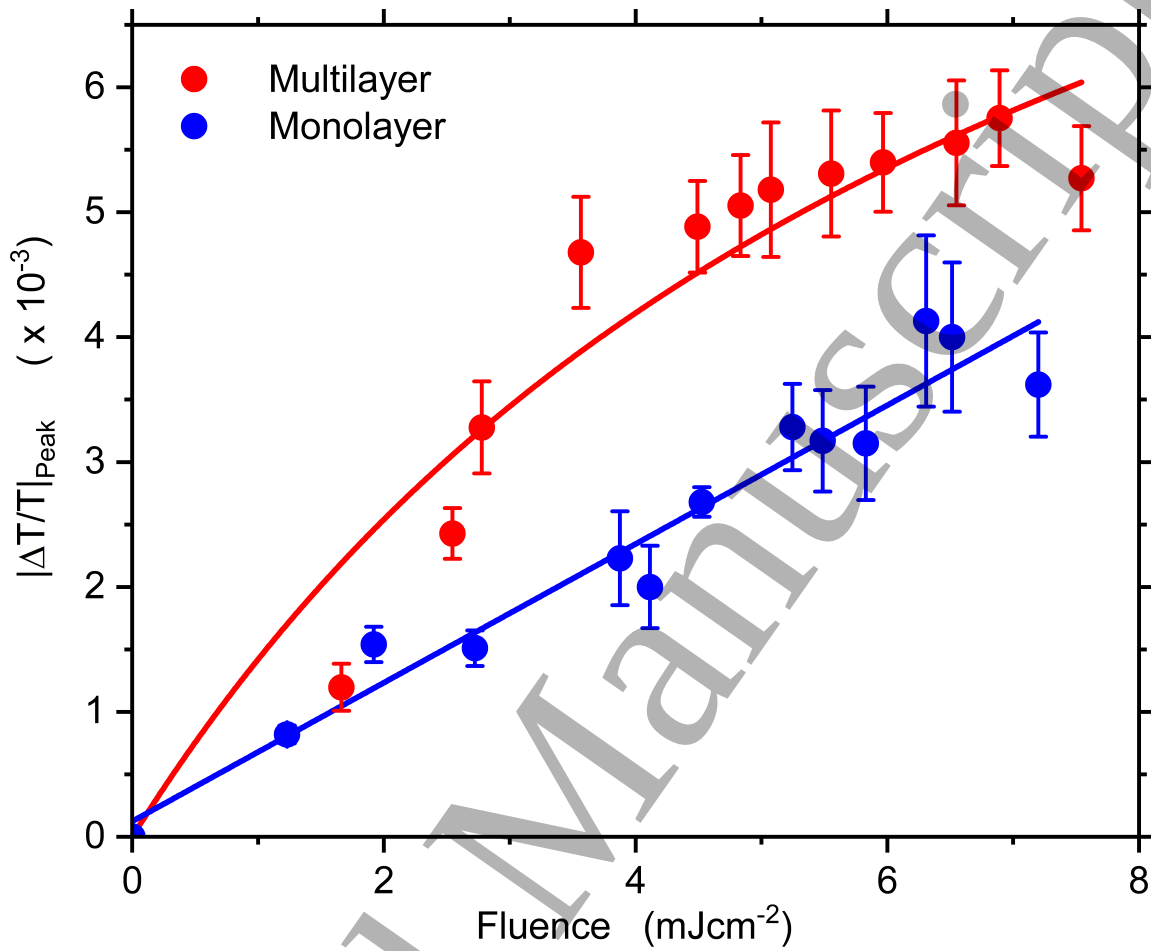


Figure 4: fluence dependence of the initial negative peak magnitude of $\Delta T/T$ for the MuL MoS₂ (red dots) and MoL MoS₂ (blue dots). The solid lines represent the best fit to the corresponding data sets.

$$\mathcal{F} = A_0 \left[1 + \mathcal{E} \left(\frac{t}{t_0} \right) \right] \exp \left(-\frac{t}{\tau_s} \right) + B_0 \left[1 - \exp \left(\frac{t - t_d}{\tau_r} \right) \right] \exp \left(-\frac{t - t_d}{\tau_L} \right). \quad (3)$$

The first part of the equation represents the initial fast negative change in the transmission and a corresponding exponential recovery with a recovery time τ_s . Here, A_0 denotes the amplitude of the initial negative transient signal, while t_0 represents its build-up time. The second term accounts for the positive $\Delta T/T$ component that emerges after the recovery of the initial negative signal. Here, τ_r describes the build-up time of this delayed positive response. The decay constant τ_L represents the subsequent relaxation of this positive component. The short-range decay time component, τ_s ,

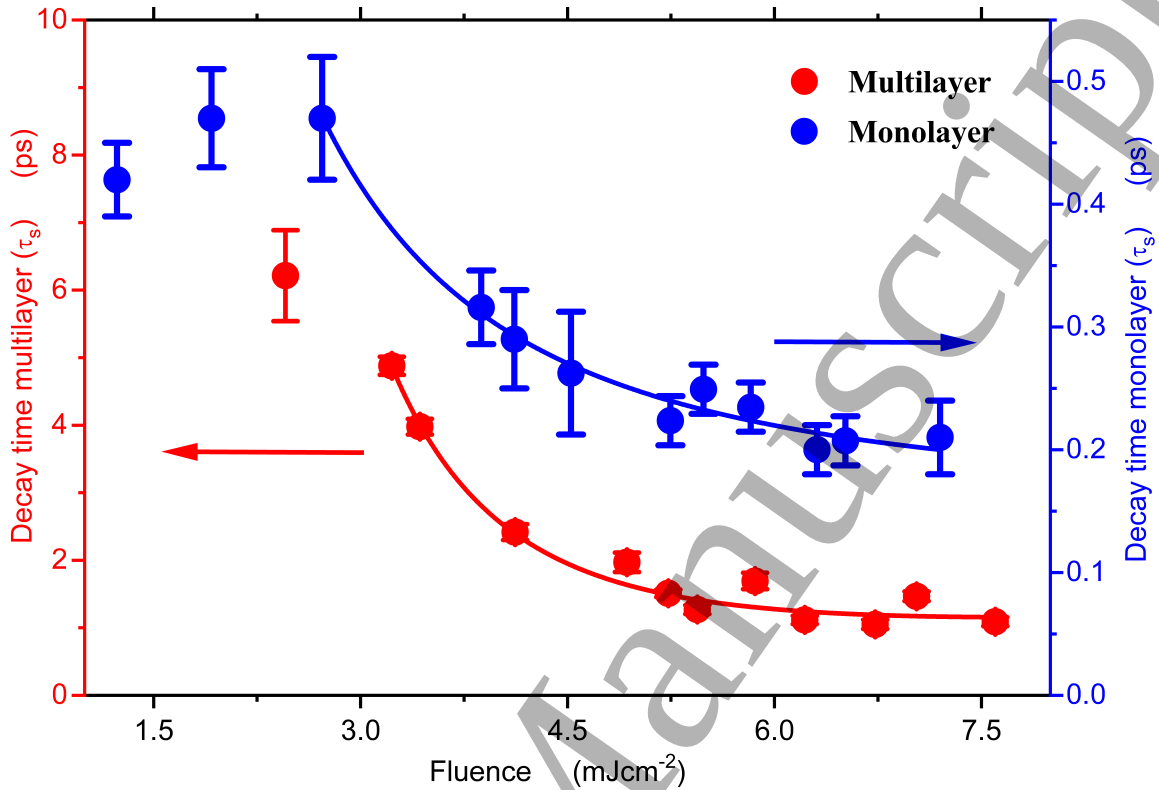


Figure 5: Pump fluence dependence of the relaxation times of the MuL (τ_{MuL} , red dots) and MoL (τ_{MoL}^s , blue dots) MoS₂ flakes. The solid lines indicate the best fits to the data.

for MoL MoS₂ is also plotted in Fig.5. At lower pump fluences ($< 2.7 \text{ mJ cm}^{-2}$), τ_s remains almost constant around 0.4 ps. However, as the pump fluence exceeds 2.7 mJ cm^{-2} , the relaxation time decreases rapidly, showing a similar trend as that of the MuL case.

Following the recovery of negative change in transmission, the MoL flake shows a positive transmission change even at a pump fluence of 3.9 mJ cm^{-2} which subsequently increases with higher pump fluences. However, the MuL flake shows only a small positive change at the highest fluence used in the present measurements (7.5 mJ cm^{-2}). Furthermore, the positive peak in $\Delta T/T$ for the MoL sample appears around 3 ps followed by a gradual decay. In contrast, for the MuL sample, the positive change in $\Delta T/T$ remains almost constant without any noticeable recovery over tens of picoseconds of delay time.

It is well established that the electronic band structure of MoS₂ is highly dependent on the number of layers in the material [1, 2]. Figure 6, panels (a) and (b), represents a simplified schematic of the relevant band structure for MoL and MuL MoS₂. In MoL MoS₂, the material exhibits a direct bandgap at the *K* valley, where both the conduction band minimum and the valence band maximum are located [1, 2]. When two layers are stacked, a conduction band minimum emerges at the *Q* point situated in between the Γ and *K* valleys, resulting in an indirect bandgap [1]. This transition

significantly reduces the photoluminescence efficiency of bilayer MoS₂ by approximately a factor of ~ 40 compared to MoL MoS₂ [1]. As the number of layers increases, the conduction band minima at the Q point continues to decrease in energy [1, 32]. Beyond 5 to 6 layers, the changes in the band structure stabilize and become largely independent of the number of layers. Thus, the MuL flake investigated in this study is an indirect bandgap semiconductor. Despite these structural changes, the excitonic resonance energy of MuL MoS₂ remains nearly identical to that in MoL MoS₂, at 1.83 eV [1, 15]. However, the A-exciton binding energy differs significantly in both cases: ~ 45 meV in MuL which is an order of magnitude lower than in MoL MoS₂ [15, 16].

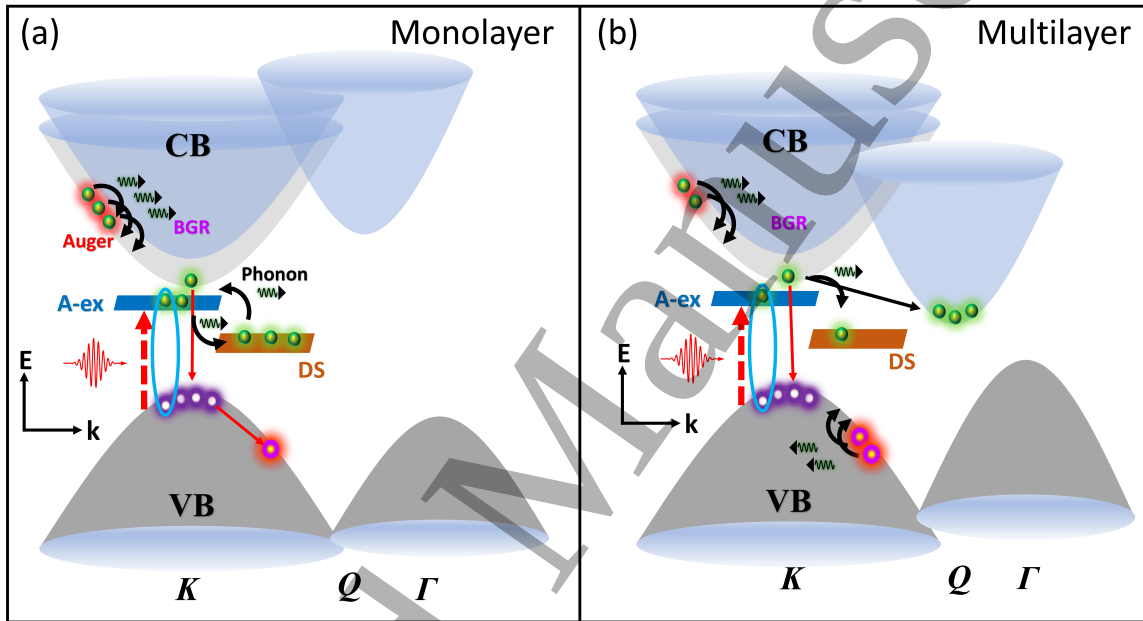


Figure 6: Simplified schematic of the band structure near the band edge and various processes that happen at femtosecond to few tens of picosecond time scale when pump photon energy is tuned to A-exciton: (a) MuL MoS₂ with exciton dissociation, Auger scattering, and intervalley scattering and (b) MoL MoS₂ exciton dissociation and Auger scattering.

Let us now examine the dynamical processes that drive the transient optical response of the MuL compared to the MoL MoS₂ following excitation by a short laser pulse. Several groups have studied carrier dynamics in MuL and MoL MoS₂ flakes at the A-exciton transition energy and reported various physical phenomena that dominate for a few hundred of picoseconds after an ultrafast pulse excitation. The observed physical phenomena are strongly dependent on the excitation photon energy, excitation density, and probing photon energy, as well as the type of samples and delay time, making understanding of the carrier dynamics quite complicated. Various processes like exciton formation [21, 39, 40], exciton dissociation [35, 37, 41, 42], exciton-exciton annihilation [17, 18, 20, 43, 44], carrier capture to defect and trap states [45–47], Auger and defect-assisted Auger scattering [32, 34, 48, 49], hot-phonon reabsorption [22, 50], and intervalley scattering [51, 52] are likely to occur in both MuL and MoL MoS₂ flakes. These processes will lead to band bleaching [53], band

gap renormalization (BGR) [50, 54], exciton peak shift and broadening [22, 43, 55], etc.

In MoL and MuL MoS₂, the exciton Mott transition typically occurs at a critical photoinduced carrier density per layer on the order of 10^{12} cm^{-2} ; specifically, Liu et al. estimated a threshold density of $\sim 4.3 \times 10^{12} \text{ cm}^{-2}$ for the onset of this transition [31]. Above this critical density, enhanced Coulomb screening by the photogenerated electron-hole gas diminishes the exciton binding energy and induces exciton ionization into a correlated electron-hole plasma. This many-body regime is characterized by a severe suppression and broadening of excitonic resonances, bleaching of the A and B excitons, and pronounced BGR. In ultrafast transient absorption spectroscopy, this high-density response manifests as a characteristic combination of exciton bleaching and derivative-like or red-shifted photoinduced absorption features. This spectral profile reflects the competing dynamics between quasi-particle bandgap shrinkage and the reduction of exciton binding energy. This interpretation is strongly supported by time-resolved angle-resolved photoemission spectroscopy (TR-ARPES) on MoS₂, which demonstrated a BGR of up to $\sim 360 \text{ meV}$ above the Mott threshold [31]. Consequently, because the estimated NPL in this work exceeds the 10^{13} cm^{-2} regime, accompanied by pronounced exciton bleaching, our observations are highly consistent with an excitation regime situated at or well beyond the exciton Mott transition. Driven by these mechanisms, the optical properties of the material will change, leading to a measurable change in the transmission of the probe beam. However, the relative contribution and temporal evolution of these mechanisms can vary significantly between the two cases due to differences in their band structures, carrier relaxation pathways, and interlayer interactions. Earlier reports by Bera et al. and Katsch et al. established that elevating carrier densities beyond the Mott limit leads to a simultaneous shift and broadening of the excitonic resonances, [56, 57]. While both phenomena inherently contribute to the modulated transmission signal, linewidth broadening is conventionally the dominant factor when exciton-exciton and exciton-phonon interactions prevail. Crucially, however, Bera et al. demonstrated that the emergence of a high-density electron-hole plasma above the Mott threshold drives bandgap renormalization, leading to reduced transmission (-ve initial peak in $\Delta T/T$ both in MoL and MuL), which is dominant in our studies, resulting in a pronounced redshift of the exciton peak. Since excitonic annihilation into an electron-hole plasma occurs rapidly at the early stages of photoexcitation, it is expected that the exciton peak shift serves as the primary mechanism governing the observed transient transmission changes, overshadowing the contribution from peak broadening.

In this study, carriers are excited at the A-exciton transition energy. At this energy, the absorption coefficient of a MoL MoS₂ is approximately 40% higher than that of MuL MoS₂ [58]. Assuming each absorbed photon generates one electron-hole pair, the carrier density in MoL is expected to be 40% higher than in MuL MoS₂. Consequently, a larger $|\Delta T/T|_{peak}$ would be anticipated for MoL. However, experimental observations reveal the opposite, with MuL exhibiting a higher $|\Delta T/T|_{peak}$ (Fig. 4). This contradiction can be explained by the lower exciton binding energy in MuL, which facilitates more efficient exciton dissociation compared to MoL [15, 41, 42]. The weaker binding energy of the exciton is coming from the multilayered structure of the MuL. This further leads to stronger dielectric screening and availability of additional intervalley scattering. In addition, in multilayer MoS₂, the optical absorption and dielectric response seen by the probe is also modified [15, 59]. This results in a higher density of free carriers near the band edge in MuL leading to stronger bandgap

renormalization (BGR), despite its lower absorption. This BGR is driven by many-body Coulomb interaction, inducing a reconfiguration of the electronic band structure and reducing the bandgap energy. This accumulation of the free carriers screens the electron-hole Coulomb attraction, causing a redshift in exciton binding energy [14, 60]. Thus, the larger $|\Delta T/T|_{Peak}$ observed in MuL is attributed to its greater exciton dissociation efficiency and stronger BGR at comparable excitation densities (Fig.4). Over time, the free carriers in MuL are expected to decay to defect states and also form hot carriers via Auger and defect-assisted Auger processes as that of MoL MoS₂ [32, 45–47]. The Auger scattering not only contributes to the relaxation of some carriers but also excites an equal number of carriers to the higher excited states (Fig.6(a)). As these carriers relax to the band edge, eventually excitons will start forming again due to lower carrier density as compared to the initial excitation condition. Such formation of the A-exciton would lead to bleaching of available states for absorption of the probe, leading to an increase in pump transmission which is stronger in MoL MoS₂ even during lower pump fluences leading to a later band bleaching with a positive peak in $\Delta T/T$. The band bleaching will give a positive contribution to the $\Delta T/T$ canceling the negative effect of BGR. Thus, the action of Auger should also be coupled with the bleaching of states later and we attribute this to the formation of excitons in the present case.

Additionally, a significant fraction of electrons relax to *Q*-valley in MuL within tens of femtoseconds, leaving holes at *K* valley [51]. These *Q*-valley electrons have lifetimes of several picoseconds [51, 52]. Although some carriers relax to defect states, a substantial population remains in the conduction band as free carriers in MuL. Moreover, since these carriers occupy different valleys, the Auger and defect-assisted Auger recombination processes are less efficient in MuL. Additionally, the greater thickness of MuL MoS₂, enhances the dielectric screening and reduces quantum confinement compared to MoL. These factors weaken the Coulombic interaction, further suppressing Auger scattering processes [56, 61]. Consequently, carrier relaxation is slower in MuL, resulting in longer recovery times for transient signals (Fig.5). For instance, at a pump fluence of 2.5 mJ cm⁻², the decay time of MuL is ~ 4.8 ps, whereas in MoL, it is ~ 0.4 ps. Similarly, at a higher fluence of 7 mJ cm⁻² pump fluence, carrier relaxation time in MuL is about 1 ps while in the case of MoL, it is 0.2 ps. Furthermore, excitation at 7.5 mJ cm⁻² pump fluence generates a high density of hot phonons in MuL MoS₂ leading to phonon bottlenecking effect due to reabsorption of the hot phonons as that of MoL which turns the BGR to BF as shown in Fig.3(a). At high excitation density, Auger-assisted relaxation and hot-carrier cooling generate a larger phonon population. These phonons are being reabsorbed by the carriers in defect states and repopulate the A-exciton state. This additional back-transfer further slows the overall decay, reinforcing the bottleneck effect, [35]. This process is an intrinsic nonequilibrium effect of the photoexcited samples under high excitation density, where a large population of carriers and emitted phonons are created within an ultrashort time window. reabsorption of hot phonons by carriers can slow carrier cooling, and the reduction in the number of available final states, including partial saturation of defect-assisted relaxation channels, can further prolong the relaxation process. However, the bottlenecking effect is more prominent in MoL and arises at significantly lower pump fluence compared to MuL. This difference is attributed to the intervalley scattering in MuL, which provides multiple relaxation pathways for carriers, unlike MoL, where excitation and relaxation of the carriers primarily occur at the *K* valley. Further, intervalley

scattering and relaxation in MuL also reduce the possibility of the formation of A-excitons. Consequently, the bleaching of $\Delta T/T$ in MuL MoS₂ is much weaker and only becomes evident at much higher excitation fluence.

3 Conclusion

In this article, we have demonstrated that the ultrafast responses of MoL and MuL MoS₂ differ significantly under excitation by ultrashort pulses. These studies were carried out at high photoexcitation densities in the range of $\sim 6.6 \times 10^{13} \text{ cm}^{-2}$ to $\sim 4.16 \times 10^{14} \text{ cm}^{-2}$. In spite of the fact that the MuL has a lower excitation density as compared to MoL, it still exhibited a higher change in $|\Delta T/T|_{Peak}$ when compared to that of MoL MoS₂ and clear saturation behavior, since the dissociation of excitons is much easier in MuL due to the lower exciton binding energy. Both MuL and MoL show Auger and defect-assisted Auger recombination and a positive change in $\Delta T/T$ at later times, but the MuL shows a much longer recovery time and a small positive change in $\Delta T/T$ at high excitation densities. Both of these changes are attributed to the carriers scattered to other valleys in the conduction band, which increases the carrier lifetime by suppressing exciton formation. The insights gained from these studies contribute to the understanding of ultrafast carrier dynamics in MoS₂ and are expected to aid in the development of optoelectronics applications involving monolayer and multilayer MoS₂, such as lasers, OPAs, and detectors.

Acknowledgments

All authors sincerely thank Mr. Vijay Singh Dawer for his assistance during the measurements. The experiments presented in this work were conducted while all authors were affiliated with the Nano Science Laboratory, Materials Science Section, Raja Ramanna Centre for Advanced Technology (RRCAT), Indore. D.P.K. and S.G. acknowledge financial support from RRCAT, Indore, under the HBNI PhD program. J.J. gratefully acknowledges funding from the Deutsche Forschungsgemeinschaft (DFG, German Research Foundation) through Project No. BO1823/12 - FOR 5249 (QUAST).

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

References

1. Mak KF, Lee C, Hone J, Shan J, and Heinz TF. Atomically thin MoS₂: a new direct-gap semiconductor. *Phys. Rev. Lett.* 2010;105:136805.
2. Splendiani A, Sun L, Zhang Y, et al. Emerging photoluminescence in monolayer MoS₂. *Nano Lett.* 2010;10:1271.
3. Lembke D and Kis A. Breakdown of high-performance monolayer MoS₂ transistors. *ACS nano* 2012;6:10070.
4. Zheng H, Yang B, Wang D, Han R, Du X, and Yan Y. Tuning magnetism of monolayer MoS₂ by doping vacancy and applying strain. *Appl. Phys. Lett.* 2014;104:132403.
5. Luo YK, Xu J, Zhu T, et al. Opto-valleytronic spin injection in monolayer MoS₂/few-layer graphene hybrid spin valves. *Nano Lett.* 2017;17:3877.
6. Kim S, Konar A, Hwang WS, et al. High-mobility and low-power thin-film transistors based on multilayer MoS₂ crystals. *Nat. Commun.* 2012;3:1011.
7. Xie Y, Zhang B, Wang S, et al. Ultrabroadband MoS₂ photodetector with spectral response from 445 to 2717 nm. *Adv. Mater.* 2017;29:1605972.
8. Singh E, Kim KS, Yeom GY, and Nalwa HS. Atomically thin-layered molybdenum disulfide (MoS₂) for bulk-heterojunction solar cells. *ACS Appl. Mater. Interfaces.* 2017;9:3223–45.
9. Singh E, Singh P, Kim KS, Yeom GY, and Nalwa HS. Flexible molybdenum disulfide (MoS₂) atomic layers for wearable electronics and optoelectronics. *ACS Appl. Mater. Interfaces.* 2019;11:11061–105.
10. Lazanas AC and Prodromidis MI. Two-dimensional inorganic nanosheets: production and utility in the development of novel electrochemical (bio) sensors and gas-sensing applications. *Microchim. Acta* 2021;188:1–34.
11. Brumme T, Calandra M, and Mauri F. First-principles theory of field-effect doping in transition-metal dichalcogenides: Structural properties, electronic structure, Hall coefficient, and electrical conductivity. *Phys. Rev. B* 2015;91:155436.
12. Yuan L, Wang T, Zhu T, Zhou M, and Huang L. Exciton dynamics, transport, and annihilation in atomically thin two-dimensional semiconductors. *J. Phys. Chem. Lett.* 2017;8:3371.
13. Chernikov A, Berkelbach TC, Hill HM, et al. Exciton binding energy and nonhydrogenic Rydberg series in monolayer WS₂. *Phys. Rev. Lett.* 2014;113:076802.
14. Ugeda MM, Bradley AJ, Shi SF, et al. Giant bandgap renormalization and excitonic effects in a monolayer transition metal dichalcogenide semiconductor. *Nat. Mater.* 2014;13:1091.
15. Borzda T, Gadermaier C, Vujcic N, et al. Charge photogeneration in few-layer MoS₂. *Adv. Funct. Mater.* 2015;25:3351.
16. Evans B and Young P. Exciton spectra in thin crystals: the diamagnetic effect. *Proc. Phys. Soc. (1958-1967)* 1967;91:475.

17. Lee KJ, Xin W, and Guo C. Annihilation mechanism of excitons in a MoS₂ monolayer through direct Förster-type energy transfer and multistep diffusion. *Phys. Rev. B* 19 2020;101:195407.

18. Tsai HS, Huang YH, Tsai PC, et al. Ultrafast Exciton Dynamics in Scalable Monolayer MoS₂ Synthesized by Metal Sulfurization. *ACS omega* 2020;5:10725.

19. Srivastava YK, Chaturvedi A, Manjappa M, et al. MoS₂ for ultrafast all-optical switching and modulation of THz fano metaphotonic devices. *Advanced Optical Materials* 2017;5:1700762.

20. Zhang T and Wang J. Defect-Enhanced Exciton–Exciton Annihilation in Monolayer Transition Metal Dichalcogenides at High Exciton Densities. *ACS Photonics* 2021.

21. Chi Z, Chen H, Zhao Q, and Weng YX. Observation of the hot-phonon effect in monolayer MoS₂. *Nanotechnology* 2020;31:235712.

22. Wang W, Sui N, Chi X, et al. Investigation of Hot Carrier Cooling Dynamics in Monolayer MoS₂. *J. Phys. Chem. Lett.* 2021;12:861.

23. Pan C, Jiang L, Sun J, et al. Ultrafast optical response and ablation mechanisms of molybdenum disulfide under intense femtosecond laser irradiation. *Light Sci. Appl.* 2020;9:1.

24. Völzer T, Fennel F, Korn T, and Lochbrunner S. Fluence-dependent dynamics of localized excited species in monolayer versus bulk MoS₂. *Phys. Rev. B* 2021;103:045423.

25. Del Águila AG, Wong YR, Wadgaonkar I, et al. Ultrafast exciton fluid flow in an atomically thin MoS₂ semiconductor. *Nat. Nanotechnol.* 2023;18:1012–9.

26. Giroto N, Caruso F, and Novko D. Ultrafast Nonadiabatic Phonon Renormalization in Photoexcited Single-Layer MoS₂. *The Journal of Physical Chemistry C* 2023;127:16515–24.

27. Xu C, Zhou G, Alexeev EM, et al. Ultrafast Electronic Relaxation Dynamics of Atomically Thin MoS₂ Is Accelerated by Wrinkling. *Acs Nano* 2023;17:16682–94.

28. Trovatello C, Marini A, Xu X, et al. Optical parametric amplification by monolayer transition metal dichalcogenides. *Nat. Photonics* 2021;15:6–10.

29. Zhao G, Hou J, Wu Y, He J, and Hao X. Preparation of 2D MoS₂/graphene heterostructure through a monolayer intercalation method and its application as an optical modulator in pulsed laser generation. *Adv. Opt. Mater.* 2015;3:937.

30. Zhou YH, An HN, Gao C, Zheng ZQ, and Wang B. UV–Vis–NIR photodetector based on monolayer MoS₂. *Mater. Lett.* 2019;237:298.

31. Liu F, Ziffer ME, Hansen KR, Wang J, and Zhu X. Direct determination of band-gap renormalization in the photoexcited monolayer MoS₂. *Phys. Rev. Lett.* 2019;122:246803.

32. Wang H, Zhang C, and Rana F. Surface recombination limited lifetimes of photoexcited carriers in few-layer transition metal dichalcogenide MoS₂. *Nano Lett.* 2015;15:8204.

33. 2DLayer. 2DLayer. <https://2dlayer.com>. Procured from 2 Davis Drive, Durham, NC 27709, USA. 2020.

34. Zhang C, Wang H, Chan W, Manolatou C, and Rana F. Absorption of light by excitons and trions in monolayers of metal dichalcogenide MoS₂: Experiments and theory. *Phys. Rev. B* 2014;89:205436.
35. Khatua DP, Singh A, Gurung S, and Jayabalan J. Excitation Density Dependent Carrier Dynamics in a Monolayer MoS₂: Exciton Dissociation, Formation and Bottle Necking. *Micro and Nanostructures* 2022;165:207205.
36. Khatua DP, Singh A, Gurung S, Tanwar M, Kumar R, and Jayabalan J. A comparative study of ultrafast carrier dynamics near A, B, and C-excitons in a monolayer MoS₂ at high excitation densities. *Opt. Mater.* 2022;126:112224.
37. Khatua DP, Singh A, Gurung S, Khan S, and Jayabalan J. Ultrafast carrier dynamics in a monolayer MoS₂ at carrier densities well above mott density. *J. Phys. Condens. Matter* 2022;34:155401.
38. Song B, Gu H, Fang M, et al. Layer-dependent dielectric function of wafer-scale 2D MoS₂. *Advanced Optical Materials* 2019;7:1801250.
39. Yu K, Wang J, Chen J, and Wang GP. Inhomogeneous photocarrier dynamics and transport in monolayer MoS₂ by ultrafast microscopy. *Nanotechnology* 2019;30:485701.
40. Ceballos F, Cui Q, Bellus MZ, and Zhao H. Exciton formation in monolayer transition metal dichalcogenides. *Nanoscale* 2016;8:11681.
41. Nie Z, Long R, Sun L, et al. Ultrafast carrier thermalization and cooling dynamics in few-layer MoS₂. *ACS nano* 2014;8:10931.
42. Nie Z, Long R, Teguh JS, et al. Ultrafast electron and hole relaxation pathways in few-layer MoS₂. *J. Phys. Chem. C* 2015;119:20698.
43. Wang L, Wang Z, Wang HY, et al. Slow cooling and efficient extraction of C-exciton hot carriers in MoS₂ monolayer. *Nat. Commun.* 2017;8:1.
44. Sun D, Rao Y, Reider GA, et al. Observation of rapid exciton–exciton annihilation in monolayer molybdenum disulfide. *Nano Lett.* 2014;14:5625.
45. Cha S, Sung JH, Sim S, et al. 1 s-intraexcitonic dynamics in monolayer MoS₂ probed by ultrafast mid-infrared spectroscopy. *Nat. Commun.* 2016;7:1.
46. Schiettecatte P, Geiregat P, and Hens Z. Ultrafast carrier dynamics in few-layer colloidal molybdenum disulfide probed by broadband transient absorption spectroscopy. *J. Phys. Chem. C* 2019;123:10571–7.
47. Seo M, Yamaguchi H, Mohite AD, et al. Ultrafast optical microscopy of single monolayer molybdenum disulfide flakes. *Sci. Rep.* 2016;6:1.
48. Berkelbach TC, Hybertsen MS, and Reichman DR. Theory of neutral and charged excitons in monolayer transition metal dichalcogenides. *Phys. Rev. B* 2013;88:045318.
49. Wang H, Zhang C, and Rana F. Ultrafast dynamics of defect-assisted electron–hole recombination in monolayer MoS₂. *Nano Lett.* 2015;15:339.

50. He X, Chebl M, and Yang DS. Cross-Examination of Ultrafast Structural, Interfacial, and Carrier Dynamics of Supported Monolayer MoS₂. *Nano Lett.* 2020;20:2026.
51. Wallauer R, Reimann J, Armbrust N, Gdde J, and Hfer U. Intervalley scattering in MoS₂ imaged by two-photon photoemission with a high-harmonic probe. *Appl. Phys. Lett.* 2016;109:162102.
52. Kumar N, He J, He D, Wang Y, and Zhao H. Charge carrier dynamics in bulk MoS₂ crystal studied by transient absorption microscopy. *J. Appl. Phys.* 2013;113:133702.
53. Chernikov A, Ruppert C, Hill HM, Rigosi AF, and Heinz TF. Population inversion and giant bandgap renormalization in atomically thin WS₂ layers. *Nat. Photonics* 2015;9:466.
54. Pogna EA, Marsili M, De Fazio D, et al. Photo-induced bandgap renormalization governs the ultrafast response of single-layer MoS₂. *ACS nano* 2016;10:1182–8.
55. Sim S, Park J, Song JG, et al. Exciton dynamics in atomically thin MoS₂: interexcitonic interaction and broadening kinetics. *Phys. Rev. B* 2013;88:075434.
56. Bera SK, Shrivastava M, Bramhachari K, et al. Atom like interaction and optically tunable giant band-gap renormalization in large-area atomically thin MoS₂. *Phys. Rev. B* 2021;104:L201404.
57. Katsch F, Selig M, and Knorr A. Exciton-scattering-induced dephasing in two-dimensional semiconductors. *Physical review letters* 2020;124:257402.
58. Castellanos-Gomez A, Quereda J, Meulen HP van der, Agrait N, and Rubio-Bollinger G. Spatially resolved optical absorption spectroscopy of single-and few-layer MoS₂ by hyperspectral imaging. *Nanotechnology* 2016;27:115705.
59. Ermolaev GA, Stebunov YV, Vyshnevyy AA, et al. Broadband optical properties of monolayer and bulk MoS₂. *npj 2D Materials and Applications* 2020;4:21.
60. Cunningham PD, Hanbicki AT, McCreary KM, and Jonker BT. Photoinduced bandgap renormalization and exciton binding energy reduction in WS₂. *ACS nano* 2017;11:12601–8.
61. Sim S, Shin HS, Lee D, et al. Opposite behavior of ultrafast dynamics of exciton shift and linewidth broadening in bilayer ReS₂. *Phys. Rev. B* 2021;103:014309.